Year 7
Mathematics

Unit 1: Analysing and Displaying Data

1.3 Two-way Tables and Bar Charts

Two-way Tables and Bar Charts

STARTER · 5 MIN

Date: _____ **Title: Two-way Tables and Bar Charts**

LEARNING OBJECTIVES

- To use two-way tables

STARTER *Answer in your book. No calculators.*

1. Next two: 0, 5, 10, 15, …
2. Next two: 20, 22, 24, 26, …
3. Next two: 0, 50, 100, 150, …
4. Work out 56 + 23

Starter — Answers

STARTER

Tick or correct in green pen

1. **20, 25**

2. **28, 30**

3. **200, 250**

4. **79**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can read information from a two-way table	☐
SILVER	I can complete missing values in a two-way table	☐
GOLD	I can interpret and explain what the data in a two-way table tells me	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Two-way Tables and Bar Charts

INTRODUCTION · 3 MIN

A two-way table shows data about two things at once.

Two-way table

Rows show one thing, columns show the other.

Find the row, then the column

Totals

Each row and each column has a total.

Row totals add up to the grand total

Dual bar chart

Two bars side by side for each category.

Good for comparing two groups

Compound bar chart

Each bar is split into parts stacked up.

Bar height = the total

Examples 1 and 2 — Reading Tables and Bar Charts

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Find the row, then the column Total = add the row

Example 1

Staff at a police station:

(a) How many female Inspectors?

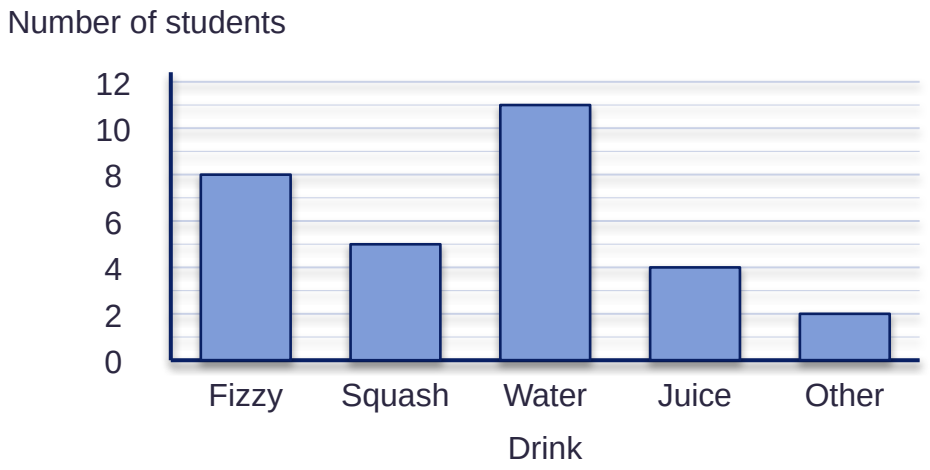
(b) How many Sergeants?

	Male	Female	Total
Constable	56	23	79
Sergeant	8	5	13
Inspector	2	4	6
Chief Insp.	1	1	2
Total	67	33	100

Example 2

Favourite drinks of a class:

How many students are there?



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Find the row, then the column

(a) Inspector row, Female column

Female Inspectors = 4

(b) Sergeant row, Total column

Sergeants = 13

Tip: the Total row and column add up.

Example 2

Add the heights of all the bars

Students = 8 + 5 + 11 + 4 + 2

= 30

Tip: read each bar against the scale.

Task 1

PRACTICE · 9 MIN

BRONZE

Use the two-way table to answer each question

Hint Find the row, then the column
Total = add the row

Lunch choices of some Year 7 students

	Hot meal	Sandwich	Salad	Total
Boys	14	9	3	26
Girls	8	10	6	24
Total	22	19	9	50

1. How many girls chose salad?
2. How many boys are there?
3. How many students chose a sandwich?
4. Which lunch was most popular with boys?
5. How many more girls than boys chose salad?

★ How many students altogether?

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

Lunch choices of some Year 7 students

	Hot meal	Sandwich	Salad	Total
Boys	14	9	3	26
Girls	8	10	6	24
Total	22	19	9	50

1. **6**
2. **26**
3. **19**
4. **Hot meal (14)**
5. **Difference = 6 – 3 = 3**

★ **50**

Examples 3 and 4 — Completing Two-way Tables

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint
Missing value = total − the others
Check: row totals = column totals

Example 3

Press-ups at the weekend:

Alan: Sat 20, Sun 22

Yolanda: Sat 20, Sun 15

Complete the two-way table.

	Sat	Sun	Total
Alan			
Yolanda			
Total			

Example 4

60 students go to the zoo or the museum.

Complete the two-way table.

	Zoo	Museum	Total
Boys	18		34
Girls		15	
Total			60

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Fill the rows, then add

Alan = 20 + 22 = 42

Yolanda = 20 + 15 = 35

Grand total = 42 + 35 = 77

	Sat	Sun	Total
Alan	20	22	42
Yolanda	20	15	35
Total	40	37	77

Tip: check 40 + 37 = 77 too.

Example 4

Missing value = total − the others

Boys, museum = 34 − 18 = 16

Girls = 60 − 34 = 26

Girls, zoo = 26 − 15 = 11

	Zoo	Museum	Total
Boys	18	16	34
Girls	11	15	26
Total	29	31	60

Tip: the zoo total is 18 + 11 = 29.

Task 2

PRACTICE · 9 MIN

SILVER

Find each missing value in the two-way table

Hint Missing value = total − the others
Check: row totals = column totals

Pets owned by some Year 7 students

	Dog	Cat	Other	Total
Boys	12	5		25
Girls		11	7	
Total	21			

1. How many boys have an 'other' pet?
2. How many students have a cat?
3. How many girls have a dog?
4. How many girls are there?
5. How many students have an 'other' pet?

★ How many students altogether? Check it two ways.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

Pets owned by some Year 7 students

	Dog	Cat	Other	Total
Boys	12	5	8	25
Girls	9	11	7	27
Total	21	16	15	52

1. **Boys, other = 25 − 12 − 5 = 8**
2. **Cat = 5 + 11 = 16**
3. **Girls, dog = 21 − 12 = 9**
4. **Girls = 9 + 11 + 7 = 27**
5. **Other = 8 + 7 = 15**

★ **Total = 25 + 27 = 52**

Check = 21 + 16 + 15 = 52

Examples 5 and 6 — Dual and Compound Bar Charts

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Dual: bars side by side Compound: bar height = total

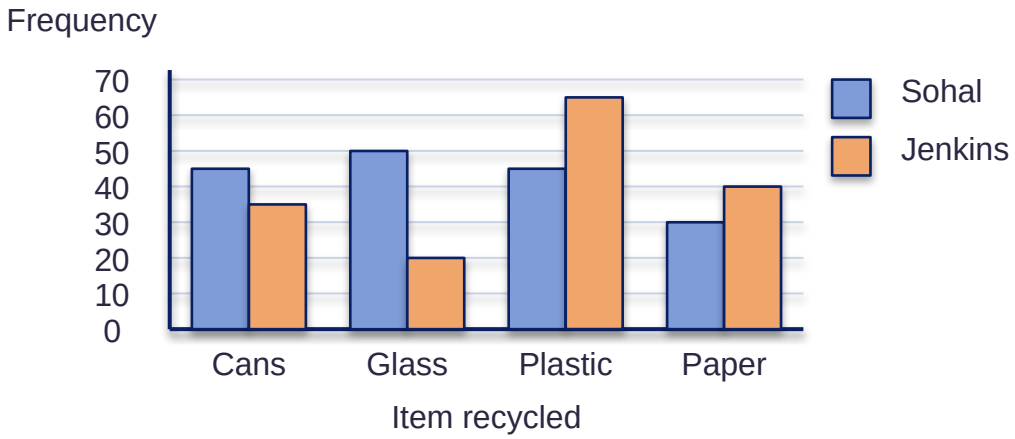
Example 5

(2 marks)

Recycling by two families:

Which family recycled more?

Show your working.

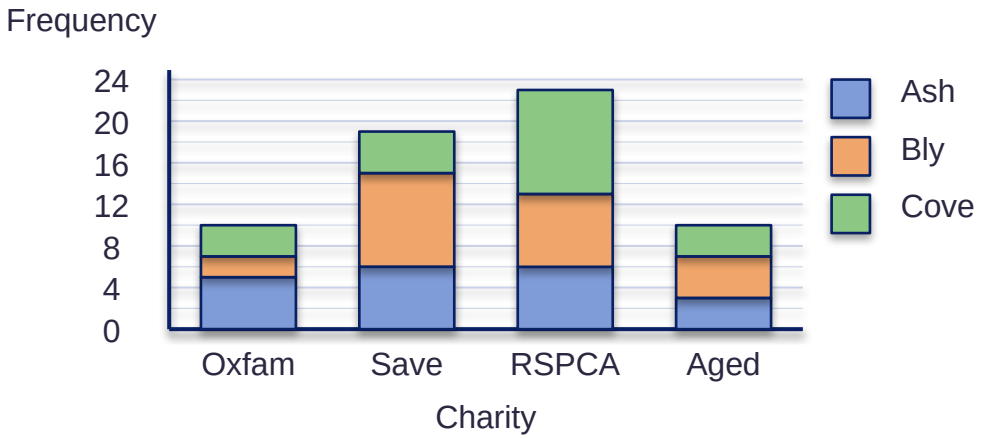


Example 6

(1 mark)

Charity chosen by people in three towns:

How many people chose RSPCA?



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Add each family's bars

Sohal = 45 + 50 + 45 + 30
= 170

Jenkins = 35 + 20 + 65 + 40
= 160

170 > 160, so the Sohal family

Tip: give both totals, then a conclusion.

Example 6

Bar height = the total

Read the top of the RSPCA bar.

RSPCA = 6 + 7 + 10
= 23 people

Cove part: 23 − 13 = 10

Tip: a part = top of part − bottom of part.

Task 3

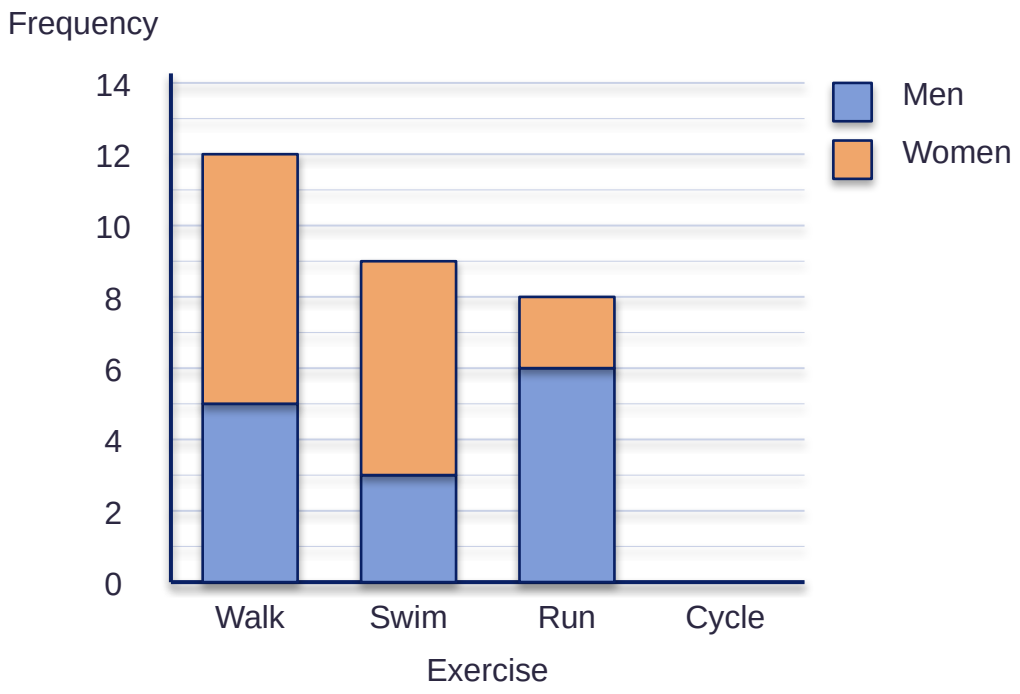
PRACTICE · 9 MIN

GOLD

Use the compound bar chart: Cycle is missing

Hint Dual: bars side by side
Compound: bar height = total

Type of exercise people do



1. How many men said they walk?
2. How many women said they swim?
3. How many people said they run?
4. 4 men and 2 women cycle. How tall is the Cycle bar?
5. Work out the total number of men.

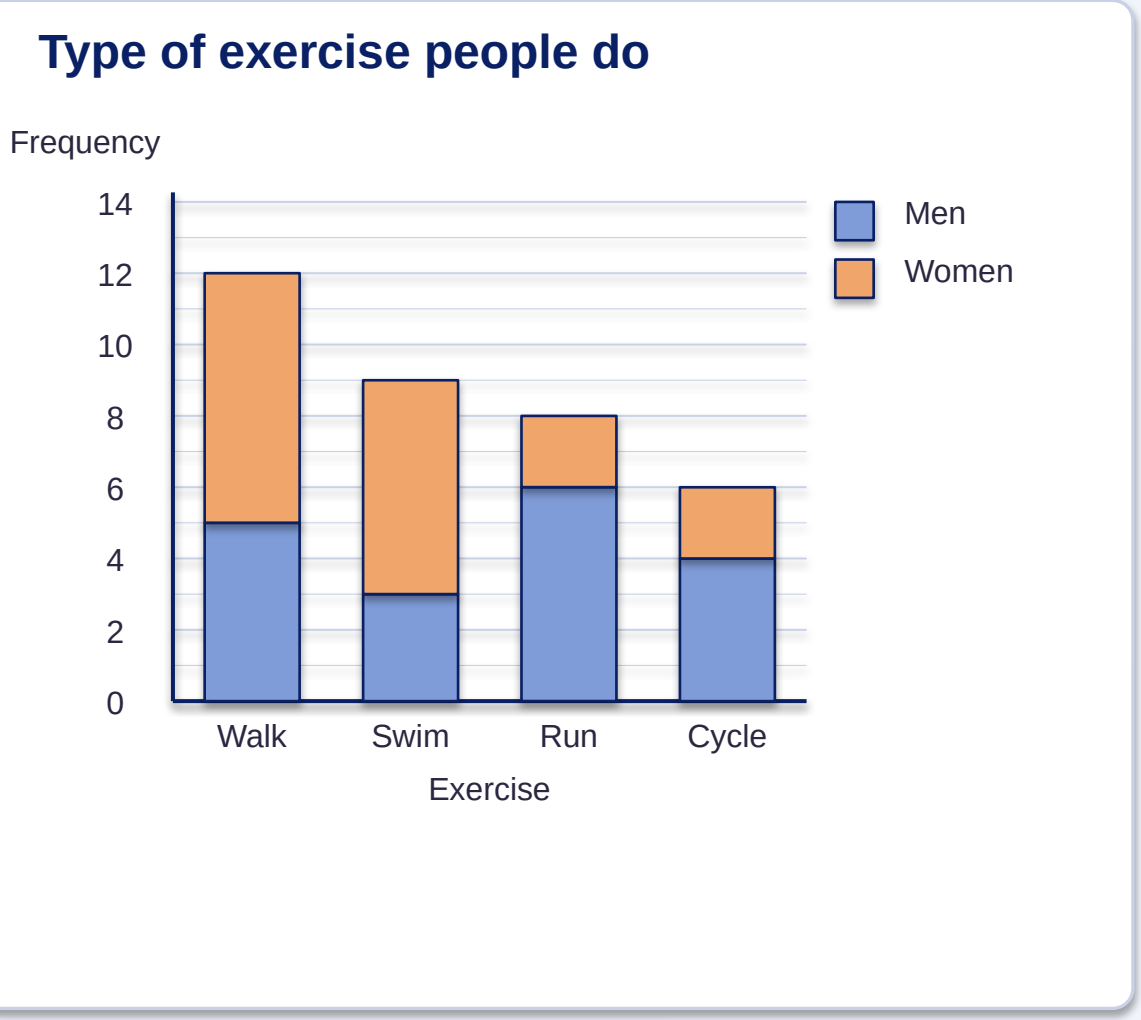
★ Were more men or women asked? Show working.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen



1. **5**
2. **6**
3. **Run = 6 + 2 = 8**
4. **6: men up to 4, women 4 to 6**
5. **Men = 5 + 3 + 6 + 4 = 18**

★ **Women = 7 + 6 + 2 + 2 = 17**
18 > 17, so more men

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Boys 25, girls 27. What is the grand total?

SILVER 25 boys: 12 have a dog, 5 a cat. How many have other pets?

GOLD When is a compound bar chart better than a dual one?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **52**

SILVER **Other = 25 − 12 − 5**

= 8

GOLD **When you want to compare the totals**

Exam Practice — Reading Tables and Charts

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Find the row, then the column
Total = add the row

Question 1

(2 marks)

The table shows police station staff.

	Male	Female	Total
Constable	56	23	79
Sergeant	8	5	13
Inspector	2	4	6
Chief Insp.	1	1	2
Total	67	33	100

(a) How many female Inspectors? (1)

.....

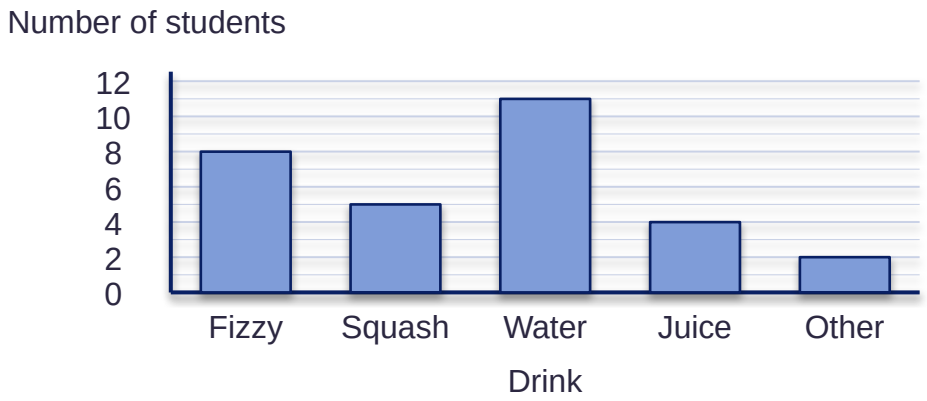
(b) How many male officers? (1)

.....

Question 2

(3 marks)

Favourite drinks of Chris's class.



(a) Which drink is the most popular? (1)

.....

(b) How many students are there? (2)

.....

Exam Practice — Two-way Tables

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint
Missing value = total − the others
Check: row totals = column totals

Question 3

(3 marks)

80 students chose French or Spanish.
45 are girls. 28 girls chose French.
20 boys chose Spanish.

	French	Spanish	Total
Boys			
Girls			
Total			80

(a) Complete the two-way table. (3)

Question 4

(3 marks)

The table shows the photos Giselle put online.

	Facebook	Flickr	Insta	Tumblr
Herself	50	30	20	20
Places	20	30	10	40

(a) How many photos are on Tumblr? (1)

.....

(b) Are there more photos of herself or of places? Show your working. (2)

.....

Exam Practice — Compound Bar Charts

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Dual: bars side by side
Compound: bar height = total

Question 5

(3 marks)

Aged: Ash 3, Bly 4, Cove 3.



(a) Complete the bar for Aged. (2)

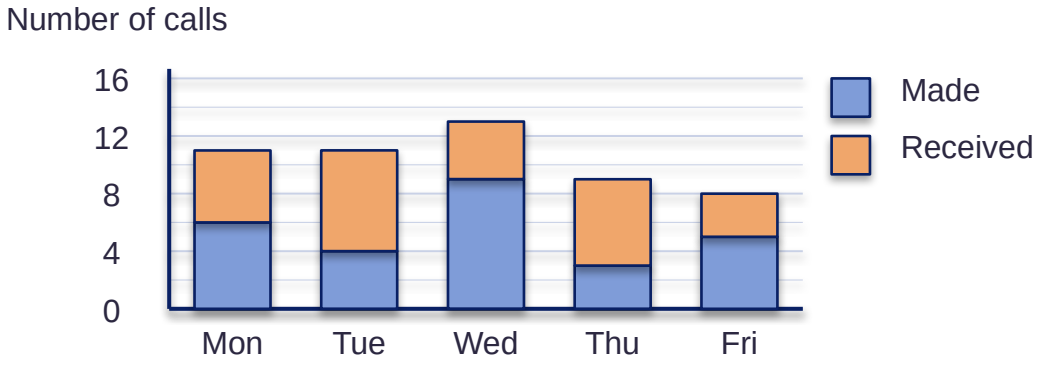
(b) How many chose Save? (1)

.....

Question 6

(3 marks)

Calls Jamila made and received.



(a) Which day had most difference? (1)

.....

(b) Did she make more calls? (2)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 4 — B1

(b) 67 — B1

Tip: read the Total row or column.

Question 2

(a) Water — B1

(b) Total = 8 + 5 + 11 + 4 + 2 — M1

= 30 — A1

Tip: read every bar against the scale.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

Boys = 80 − 45 = 35

Boys, French = 35 − 20 = 15

Girls, Spanish = 45 − 28 = 17

French 43, Spanish 37 B3

B2 for 3 or 4 correct, B1 for 2

Tip: check 43 + 37 = 80.

Question 4

(a) Tumblr = 20 + 40

= 60 B1

(b) 50 + 30 + 20 + 20 = 120 M1

20 + 30 + 10 + 40 = 100

More of herself: 120 > 100 A1

Tip: work out both totals before you decide.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) Bar to 10: 0–3, 3–7, 7–10 B2

B1 for one part correct

(b) Save = 6 + 9 + 4

= 19 B1

Tip: stack each part on top of the last.

Question 6

(a) Wednesday (9 and 4) B1

(b) Made = 6 + 4 + 9 + 3 + 5 M1

= 27

Received = 5 + 7 + 4 + 6 + 3

= 25

Yes: 27 > 25 A1

Tip: received part = top − bottom.